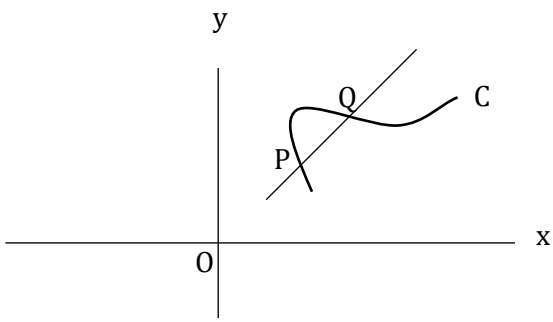
Lecture 1 & 2

Introduction to Limits

We know that an arbitrary function is defined by $y = f(x)$. If we draw the graph of the function in graph paper, that drawing is called the curve of the function.

- Given a function *f* and a point P on its graph, what would be the equation of the line tangent to the graph at P be? This question gives rise to differential calculus.

Secant Line:



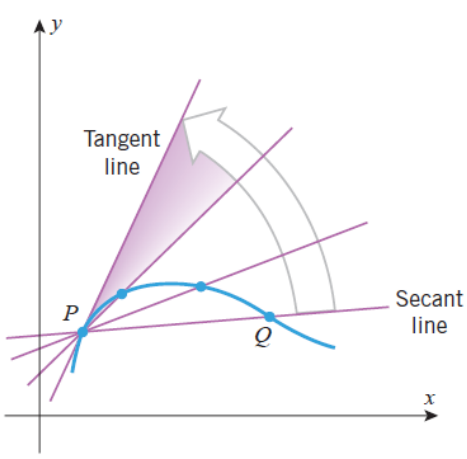
Let C be a curve on the xy-plane and P be a point on the curve.

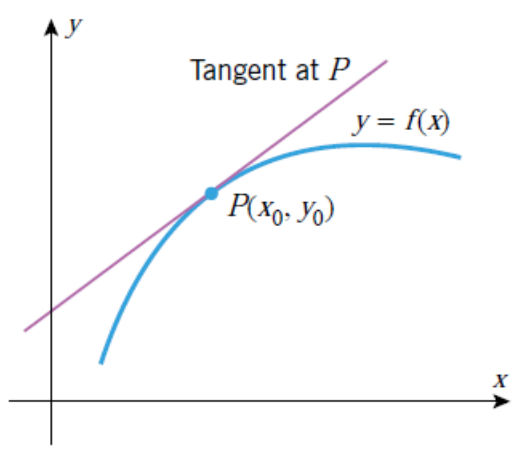
Let Q be any other point on the curve other than P.

The line through P and Q is a secant line of the curve C at P.

Tangent Line:

Consider the above secant line. If we move the point Q along C towards P, the secant line rotates. As Q approaches P, the point eventually reaches a limiting position, i.e., P and Q are infinitely close but aren't touching. The secant line through these infinitely close points is called tangent line.





Calculating the Tangent:

Consider the parabola, $y = x^2$ (It is a quadratic function). Find the tangent to this curve at point $P\ (1, 1)$. (It has basically asked to find out the equation of a straight line.)

Solution: The equation of a straight line passing through a point (x_1, y_1) and with gradient or slope = m, is given by:

$$y - y_1 = m(x - x_1) \text{_____ (i)}$$

Let, the gradient of the tangent line be m_{tan} and gradient of a secant line through $P\ (1, 1)$ and $Q(x, y)$ or $Q(x, x^2)$ be m_{sec}.

We know, the gradient of a line passing through the points (x_1, y_1) and (x_2, y_2) is given by,

$$\frac{y_2 - y_1}{x_2 - x_1}$$

So, $$m_{sec} = \frac{y_2 - y_1}{x_2 - x_1} \text{_____ (ii)}$$

Now, as Q → P , $m_{sec} \to m_{tan}$.

Or, as $x \to 1$ (algebraically equivalent to writing Q → P), $m_{sec} \to m_{tan}$.

But denominator of m_{sec} becomes 0 of if $x = 1$.

So, from equation (ii),

$$m_{sec} = \frac{x^2 - 1}{x - 1} = \frac{(x + 1)(x - 1)}{(x - 1)} = (x + 1)$$

(we can cancel out $(x - 1)$ as $x \neq 1$.)

Now, as $x \to 1$, $m_{sec} \to 1 + 1 = 2$.

So, $m_{tan} = 2$, and from equation (i),

$$y - 1 = 2(x - 1)$$

$$\text{Or}, y = 2x - 1.$$

Notice that in the above math, we found the **limiting value** of m_{sec} as $x \to 1$.

Numerical Example of Limit:

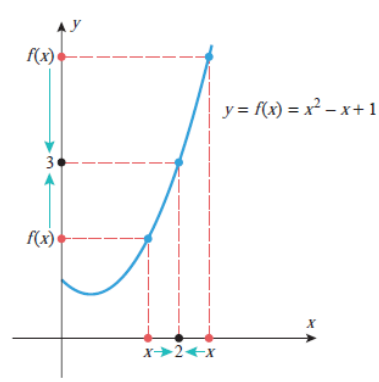
Let, $f(x) = x^2 - x + 1$. What is the value of $f(x)$ as $x \to 2$.

When $x = 1.95, f(x) = 2.852500$.

When $x = 1.995, f(x) = 2.985025$.

When $x = 1.999, f(x) = 2.997001$.

When $x = 2.05, f(x) = 3.152500$.



$$\text{When } x = 2.005, f(x) = 3.015025.$$

$$\text{When } x = 2.001, f(x) = 3.003001.$$

$$\text{So, } f(x) \to 3 \text{ as } x \to 2.$$

We say that the **limit** of $f(x) = x^2 - x + 1$ is 3 as x approaches 2. And the expression is written as,

$$\lim_{x \to 2}(x^2 - x + 1) = 3.$$

It is called Two sided limit.

<u>Limits</u>

<u>Definition:</u>

If the value of $f(x)$ can be made as close as we like to a value $'L'$ by taking values of x sufficiently close to (but not equal to) a value $'a'$ from both sides of $'a'$, then $'L'$ is called the limit of $f(x)$. And we write:

$$\lim_{x \to a} f(x) = L \text{ _____(i)}$$

Which is read as "the limit of $f(x)$ as x approaches a is L", or "$f(x)$ approaches L as x approaches a".

The expression in (i) can also be written as:

$$f(x) \to L \text{ as } x \to a$$

The limit in (i) is called a two sided limit, as it requires the values of $f(x)$ to get closer and closer to L as values of x are taken from either side of $x = a$.

<u>One-sided Limits:</u>

<u>Left Hand Limit (L.H.L):</u>

If the values of $f(x)$ can be made as close as we like to a value $'L'$ by taking values of x sufficiently close to (but less than) a value $'a'$ from the left side of $'a'$, then $'L'$ is called the left hand limit of $f(x)$. And we write:

$$\lim_{x \to a^-} f(x) = L$$

which is read as "the limit of $f(x)$ as x approaches a from the left is L".

<u>Right Hand Limit (R.H.L):</u>

If the values of $f(x)$ can be made as close as we like to a value $'L'$ by taking values of x sufficiently close to (but greater than) a value $'a'$ from the right side of $'a'$, then $'L'$ is called the right hand limit of $f(x)$. And we write:

$$\lim_{x \to a^+} f(x) = L$$

which is read as "the limit of $f(x)$ as x approaches a from the right is L".

<u>The relationship between one-sided and two-sided limits</u>:

The two-sided limit of a function $f(x)$ exists at a if and only if both of the one-sided limits exist at a and have the same value. That is, $\boldsymbol{\lim_{x\to a} f(x) = L}$ **, if and only if** $\boldsymbol{\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L}$**.**

<u>Calculating Limits:</u>

- Given the following functions,

$$f(x) = x^2 - 4x + 3$$

$$g(x) = \sqrt{2 + \mathrm{x}}$$

$$h(x) = \frac{5x^3 + 4}{x - 3}$$

Find $\lim_{x\to 2}([f(x) + g(x) - h(x)])$.

Solution:

$$\lim_{x\to 2}([f(x) + g(x) - h(x)])$$

$$= \lim_{x\to 2} f(x) + \lim_{x\to 2} g(x) - \lim_{x\to 2} h(x)$$

$$= \lim_{x\to 2}(x^2 - 4x + 3) + \lim_{x\to 2}\left(\sqrt{2 + \mathrm{x}}\right) + \lim_{x\to 2}\left(\frac{5x^3 + 4}{x - 3}\right)$$

$$= \lim_{x\to 2}(x^2) + \lim_{x\to 2}(4x) + \lim_{x\to 2}(3) + \lim_{x\to 2}\left(\sqrt{2 + \mathrm{x}}\right) - \frac{\lim_{x\to 2}(5x^3 + 4)}{\lim_{x\to 2}(x - 3)}$$

$$= 2^2 - 4(2) + 3 + \sqrt{2 + 2} - \frac{5 \times 2^3 + 4}{2 - 3} = 4 - 8 + 3 + 2 + 44$$

$$= 45$$

- Let's consider the function:

$$f(x) = \frac{2x + 8}{x^2 + x - 12}$$

Here,

$$\lim_{x\to -4} f(x) = \lim_{x\to -4} \frac{2x + 8}{x^2 + x - 12} = \frac{2(-4) + 8}{(-4)^2 + (-4) - 12} = \frac{-8 + 8}{16 - 4 - 12} = \frac{0}{0}$$

This $\frac{0}{0}$ cannot be evaluated. This isn't exactly an undefined value. It's called an indeterminate form. A form where we still can determine whether the limit exists or not.

Indeterminate Form:

Given, $f(x) = \frac{P(x)}{Q(x)}$.

- If $\lim_{x\to a} P(x) = 0$ and $\lim_{x\to a} Q(x) = 0$, then,
$$\lim_{x\to a} f(x) = \frac{0}{0}$$
- If $\lim_{x\to a} P(x) = \infty$ and $\lim_{x\to a} Q(x) = \infty$, then,
$$\lim_{x\to a} f(x) = \frac{\infty}{\infty}$$

List of Indeterminate Forms:

$$① \frac{0}{0} \quad ② \frac{\infty}{\infty} \quad ③ \, 0 \cdot \infty \quad ④ \, \infty - \infty \quad ⑤ \, 0^0 \quad ⑥ \, 1^\infty \quad ⑦ \, \infty^0$$

In the above cases, limit may exist. At times we just need to perform certain operations to obtain the limits.

Before calculating a limit always substitute the limit point of x into $f(x)$ to check for indeterminate form. i.e., for $\lim_{x\to a} f(x)$, check if f(a) gives an indeterminate form.

Example:

$$\lim_{x\to -4} \frac{2x+8}{x^2+x-12}$$
$$= \lim_{x\to -4} \frac{2(x+4)}{x^2+4x-3x-12}$$
$$= \lim_{x\to -4} \frac{2(x+4)}{x(x+4)-3(x+4)}$$
$$= \lim_{x\to -4} \frac{2(x+4)}{(x-3)(x+4)}$$
$$= \lim_{x\to -4} \frac{2}{x-3}$$
$$= \frac{2}{-4-3}$$
$$= -\frac{2}{7}$$

Brain storming:

Why $\frac{8}{\infty} \neq 0$?

Why $\frac{8}{0} \neq \infty$?

Write the correct expressions.

H.W. $\lim_{x\to 5} \frac{2x^2+9x-5}{x^2-25}$ Ans: $\frac{11}{10}$

H.W. Exercise 1.2 (3-14).

Example: Conjugate method to solve limits: (multiply by conjugate of denominator.)

$$\lim_{x\to 0} \frac{x}{\sqrt{x+1}-1}$$

$$= \lim_{x\to 0} \frac{x(\sqrt{x+1}+1)}{(\sqrt{x+1}-1)(\sqrt{x+1}+1)}$$

$$= \lim_{x\to 0} \frac{x(\sqrt{x+1}+1)}{(\sqrt{x+1})^2-1^2}$$

$$= \lim_{x\to 0} \frac{x(\sqrt{x+1}+1)}{x+1-1}$$

$$= \lim_{x\to 0} \sqrt{x+1}+1$$

$$= \sqrt{0+1}+1$$

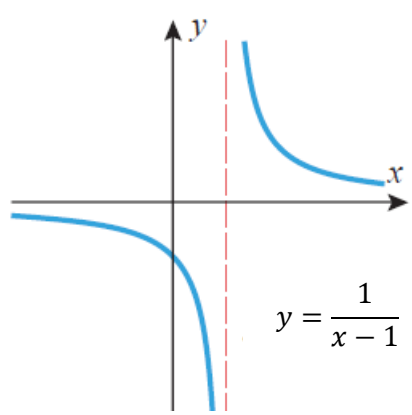
$$= 2$$

H.W: Exercise 1.2 (29,30).

- For rational functions, $f(x) = \frac{P(x)}{Q(x)}$, there are times when limit does not exist.

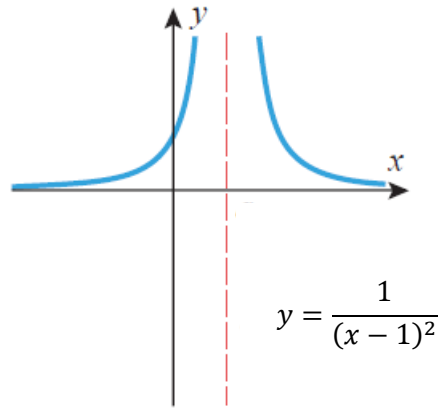
Let, $f(x) = \frac{1}{x-1}$ and $g(x) = \frac{1}{(x-1)^2}$.

The limit of $f(x) \to +\infty$ when $x \to 1^+$and $f(x) \to -\infty$ when $x \to 1^-$. As the left hand limit and the right hand limit are not equal, so, the limit does not exist.



$x = 1$

The limit of $g(x) \to +\infty$ when $x \to 1$ from both sides. So, the limit does not exist in finite number but it diverges to positive infinity so this is infinite limit.



$x = 1$

1.1.4 INFINITE LIMITS (AN INFORMAL VIEW) The expressions

$$\lim_{x \to a^-} f(x) = +\infty \quad \text{and} \quad \lim_{x \to a^+} f(x) = +\infty$$

denote that $f(x)$ increases without bound as x approaches a from the left and from the right, respectively. If both are true, then we write

$$\lim_{x \to a} f(x) = +\infty$$

Similarly, the expressions

$$\lim_{x \to a^-} f(x) = -\infty \quad \text{and} \quad \lim_{x \to a^+} f(x) = -\infty$$

denote that $f(x)$ decreases without bound as x approaches a from the left and from the right, respectively. If both are true, then we write

$$\lim_{x \to a} f(x) = -\infty$$

1.2.4 THEOREM *Let*

$$f(x) = \frac{p(x)}{q(x)}$$

be a rational function, and let a be any real number.

(*a*) *If $q(a) \neq 0$, then $\lim_{x \to a} f(x) = f(a)$.*

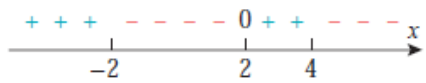
(*b*) *If $q(a) = 0$ but $p(a) \neq 0$, then $\lim_{x \to a} f(x)$ does not exist.*

$\left(\frac{c}{0} \text{ form} = \text{Limit does not exist.}\right)$

▶ **Example 8** Find

(a) $\lim_{x \to 4^+} \frac{2-x}{(x-4)(x+2)}$ (b) $\lim_{x \to 4^-} \frac{2-x}{(x-4)(x+2)}$ (c) $\lim_{x \to 4} \frac{2-x}{(x-4)(x+2)}$

Solution. In all three parts the limit of the numerator is -2, and the limit of the denominator is 0, so the limit of the ratio does not exist. To be more specific than this, we need



Sign of $\frac{2-x}{(x-4)(x+2)}$

to analyze the sign of the ratio. The sign of the ratio is determined by the signs of $2 - x$, $x - 4$ and $x + 2$.

It follows from this figure that as x approaches 4 from the right, the ratio is always negative; and as x approaches 4 from the left, the ratio is eventually positive. Thus,

$$\lim_{x \to 4^+} \frac{2-x}{(x-4)(x+2)} = -\infty \quad \text{and} \quad \lim_{x \to 4^-} \frac{2-x}{(x-4)(x+2)} = +\infty$$

Because the one-sided limits have opposite signs, all we can say about the two-sided limit is that **it does not exist**.

(c) $\lim_{x \to 5} \frac{x^2 - 3x - 10}{x^2 - 10x + 25}$

Solution (c). The numerator and the denominator both have a zero at $x = 5$, so there is a common factor of $x - 5$. Then

$$\lim_{x \to 5} \frac{x^2 - 3x - 10}{x^2 - 10x + 25} = \lim_{x \to 5} \frac{(x-5)(x+2)}{(x-5)(x-5)} = \lim_{x \to 5} \frac{x+2}{x-5}$$

However,

$$\lim_{x \to 5} (x+2) = 7 \neq 0 \quad \text{and} \quad \lim_{x \to 5} (x-5) = 0$$

so

$$\lim_{x \to 5} \frac{x^2 - 3x - 10}{x^2 - 10x + 25} = \lim_{x \to 5} \frac{x+2}{x-5}$$

does not exist.

More precisely, the sign analysis below implies that,

Sign of $\dfrac{x+2}{x-5}$

$$\lim_{x\to 5^+}\frac{x^2-3x-10}{x^2-10x+25}=\lim_{x\to 5^+}\frac{x+2}{x-5}=+\infty$$

and

$$\lim_{x\to 5^-}\frac{x^2-3x-10}{x^2-10x+25}=\lim_{x\to 5^-}\frac{x+2}{x-5}=-\infty \blacktriangleleft$$

H.W: Exercise 1.2 (15-26).

<u>Some Common Forms of Rational Functions:</u>

- $\frac{\infty}{c}$: if $+c$ then $+\infty$ and if $-c$ then $-\infty$.
- $\frac{\infty}{0}$: Use sign rule.
- $\frac{c}{\infty}$: $L=0$.
- $\frac{0}{c}$: $L=0$.
- $\frac{0}{\infty}$: $L=0$.

1.3.1 LIMITS AT INFINITY (AN INFORMAL VIEW) If the values of $f(x)$ eventually get as close as we like to a number L as x increases without bound, then we write

$$\lim_{x\to+\infty} f(x)=L \quad \text{or} \quad f(x)\to L \text{ as } x\to+\infty \qquad ($$

Similarly, if the values of $f(x)$ eventually get as close as we like to a number L as x decreases without bound, then we write

$$\lim_{x\to-\infty} f(x)=L \quad \text{or} \quad f(x)\to L \text{ as } x\to-\infty \qquad ($$

<u>Laws for Limits at Infinity:</u>

1. $\lim\limits_{x\to+\infty}\left(1+\frac{1}{x}\right)^x = e$; $\lim\limits_{x\to-\infty}\left(1+\frac{1}{x}\right)^x = e$

2. $\lim\limits_{x\to+\infty}\left(f(x)\right)^n = \left(\lim\limits_{x\to+\infty} f(x)\right)^n$; $\lim\limits_{x\to-\infty}\left(f(x)\right)^n = \left(\lim\limits_{x\to-\infty} f(x)\right)^n$

3. $\lim\limits_{x\to+\infty} Kf(x) = K\lim\limits_{x\to+\infty} f(x)$; $\lim\limits_{x\to-\infty} Kf(x) = K\lim\limits_{x\to-\infty} f(x)$ [K is a constant]

4. $\lim\limits_{x\to+\infty} K = K$; $\lim\limits_{x\to-\infty} K = K$

1.3.2 INFINITE LIMITS AT INFINITY (AN INFORMAL VIEW) If the values of $f(x)$ increase without bound as $x \to +\infty$ or as $x \to -\infty$, then we write

$$\lim_{x \to +\infty} f(x) = +\infty \quad \text{or} \quad \lim_{x \to -\infty} f(x) = +\infty$$

as appropriate; and if the values of $f(x)$ decrease without bound as $x \to +\infty$ or as $x \to -\infty$, then we write

$$\lim_{x \to +\infty} f(x) = -\infty \quad \text{or} \quad \lim_{x \to -\infty} f(x) = -\infty$$

as appropriate.

Example:

$$\lim_{x\to\infty}\left(1+\frac{1}{2x}\right)^{x} = \lim_{x\to\infty}\left[\left(1+\frac{1}{2x}\right)^{2x}\right]^{\frac{1}{2}}$$

$$= \left[\lim_{x\to\infty}\left(1+\frac{1}{2x}\right)^{2x}\right]^{\frac{1}{2}}$$

$$= e^{\frac{1}{2}} \qquad \left[\text{Since } \lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x} = e\right]$$

$$= \sqrt{e}$$

$$\boxed{e = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots}$$

Brain storming:

The correct expressions:

$\lim_{x\to\infty}\left(\frac{8}{x}\right) = 0$

$\lim_{x\to 0}\left(\frac{8}{x}\right) = \pm\infty$ /Depending on the direction

$\frac{8}{0} =$ undefined.

$\frac{8}{\infty} =$ undefined.